elements which have nearly same atomic radius as that of corresponding 4d elements. This is due to lanthanoide contraction which is to be discussed later in this unit under inner transition elements.

4.3.3 Ionization enthalpy:

Ionization energy of transition element is intermediate between those of s and p block elements. As we move from left to right in a transition metal series, the ionization enthalpy increases as expected. This is due to increase in nuclear charge corresponding to the filling of d electrons. The following figure show the trends in ionisation enthalpy of transition elements.

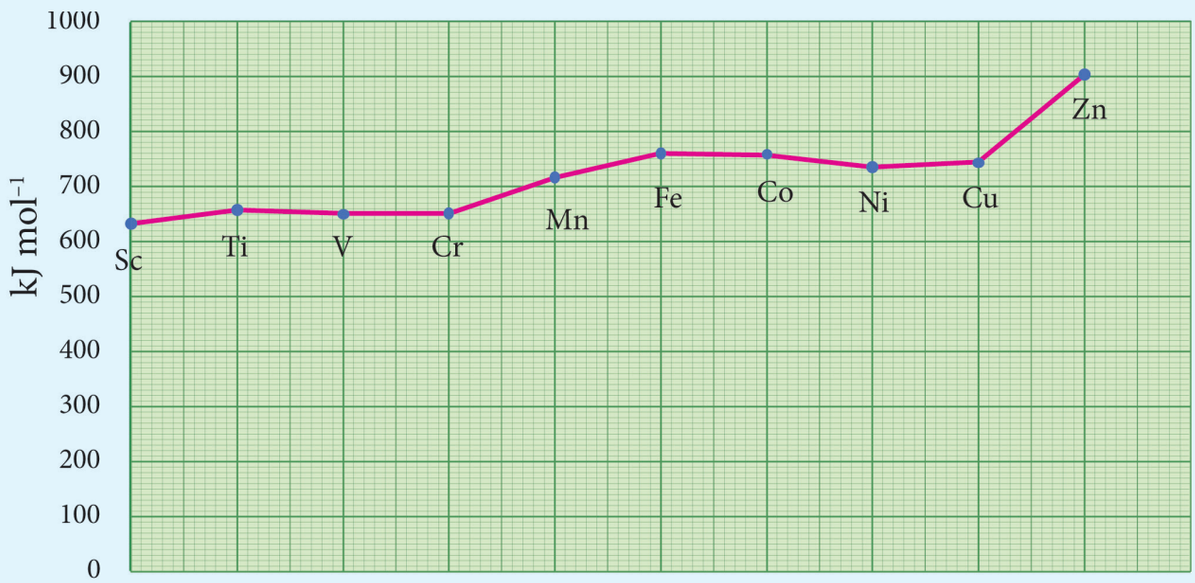


Figure 4.5 Variation in ionisation energy of 3d series elements

The increase in first ionisation enthalpy with increase in atomic number along a particular series is not regular. The added electron enters (n-1)d orbital and the inner electrons act as a shield and decrease the effect of nuclear charge on valence ns electrons. Therefore, it leads to variation in the ionization energy values.

The ionisation enthalpy values can be used to predict the thermodynamic stability of their compounds. Let us compare the ionisation energy required to form Ni^{2+} and Pt^{2+} ions.

For Nickel, $IE_1 + IE_2 = (737 + 1753)$

$= 2490\ kJmol^{-1}$

For Platinum, $IE_1 + IE_2 = (864 + 1791)$

$= 2655\ kJmol^{-1}$

Since, the energy required to form Ni^{2+} is less than that of Pt^{2+}, Ni(II) compounds are thermodynamically more stable than Pt(II) compounds.

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